

italian-creek-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types =
FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

italian_creek <- compiled_isotope |>
  filter(location == "it")
```

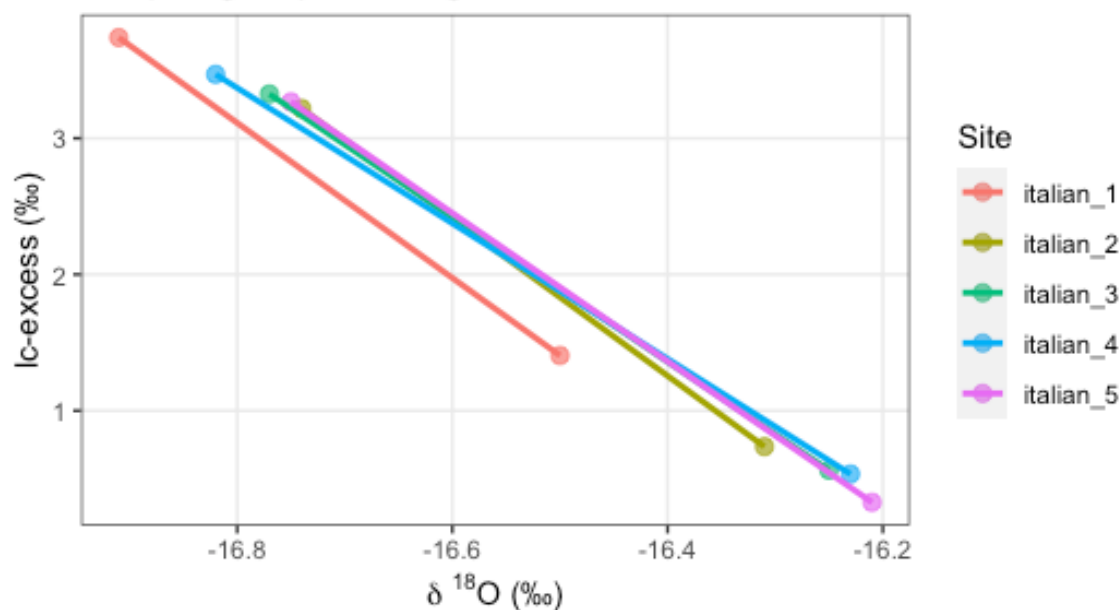
Figures

```
ggplot(data = italian_creek, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
    subtitle = "Comparing evaporation signals across Italian Creek sites",
    x = expression(paste(delta, " ^18, "0 (\u2030)")),
    y = "lc-excess (\u2030)",
    color = "Site") +
  theme_bw() +
```

```
theme(plot.title = element_text(face = "bold", size = 14),
      legend.position = "right",
      panel.grid.minor = element_blank())
```

Line-Conditioned Excess (lc-excess) vs. Delta 18O

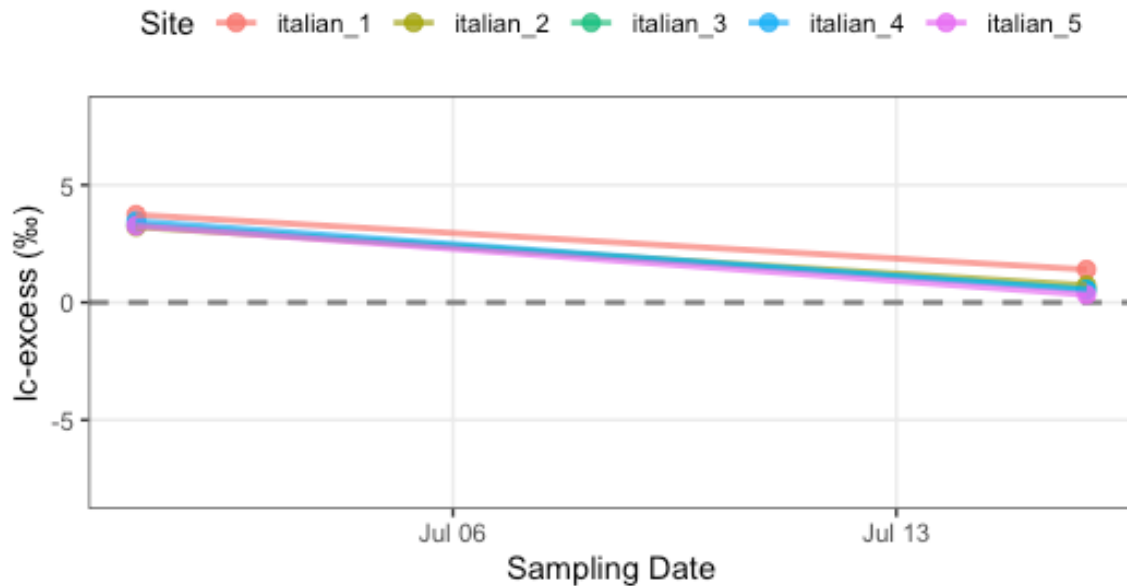
Comparing evaporation signals across Italian Creek sites



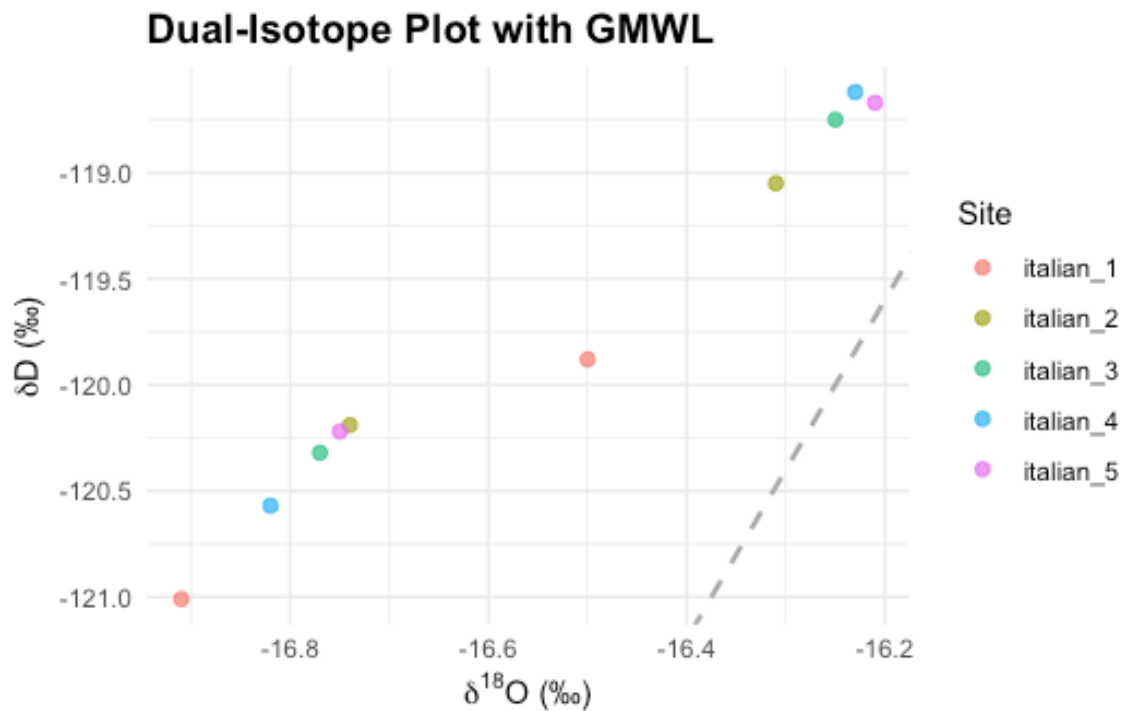
```
ggplot(data = italian_creek,
      aes(x = date, y = lc_excess, color = site, group = site)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50", linewidth =
0.8) +
  geom_point(size = 2.5, alpha = 0.8) +
  geom_line(linewidth = 1, alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
      subtitle = "Italian Creek - values below 0 indicate potential evaporative
enrichment",
      x = "Sampling Date",
      y = "lc-excess (\u2030)",
      color = "Site") +
  ylim(-8, 8) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")
```

Time-Series of Line-Conditioned Excess (lc-excess)

Italian Creek - values below 0 indicate potential evaporative enrichment

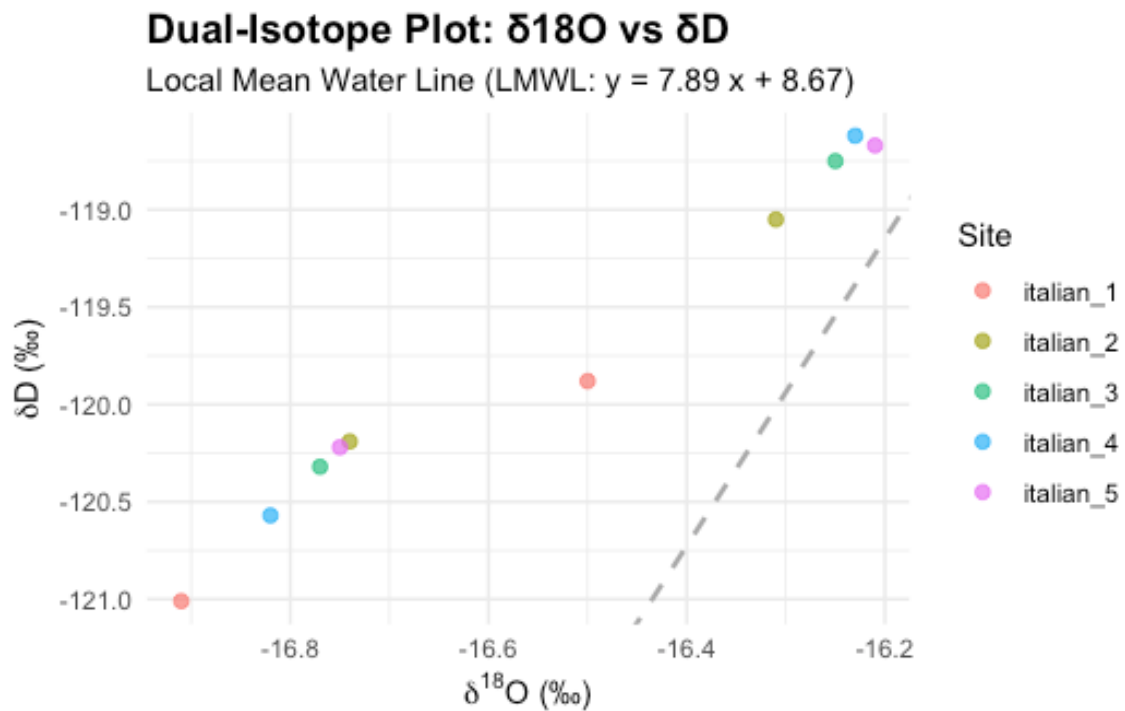


```
ggplot(italian_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Global Meteoric Water Line (GMWL: y = 8x + 10)
  geom_abline(intercept = 10, slope = 8, linetype = "dashed", color = "darkgray",
    linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot with GMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



```
# Italian Creek LMWL calculated from OPIC  $y = 7.89 x + 8.67$ 
lmwl_slope_it <- 7.89
lmwl_intercept_it <- 8.67

ggplot(italian_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = lmwl_intercept_it, slope = lmwl_slope_it,
             linetype = "dashed", color = "darkgray", linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot:  $\delta^{18}\text{O}$  vs  $\delta\text{D}$ ",
    subtitle = "Local Mean Water Line (LMWL:  $y = 7.89 x + 8.67$ )",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



```
delta_summary_it <- italian_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

ggplot(delta_summary_it, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
               width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G") +
  theme_minimal() +
  labs(title = "Mean LC Excess by Site with Standard Error",
       x = "Sampling Site",
       y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```

